THE NAKUJA PROJECT | AEROSPACE PROPULSION R&D

Comprehensive Technical Report: Characterization, Formulation, and Application Constraints of Combustion Primers for KNSB Propellant Grains

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1. DOCUMENT SCOPE AND COMBUSTION PRIMING OBJECTIVES

In solid rocket motor propulsion, achieving reliable and instantaneous ignition across all initially exposed propellant surfaces is a paramount parameter required to establish design chamber pressure and thrust profiles. While potassium nitrate-sucrose (KNSU) and potassium nitrate-dextrose (KNDX) formulations ignite with relative ease due to natural caramelization and low-temperature decomposition during heating, potassium nitrate-sorbitol (KNSB) propellants exhibit significantly higher thermal inertia and do not readily decompose or caramelize under direct flame exposure. Ad-hoc ambient testing reveals that unprimed KNSB grains demand prolonged, continuous heat flux to initiate self-sustaining combustion, introducing unacceptable ignition delays or transient "chuffing" within the motor casing.

This technical report establishes the physical, chemical, and ballistic criteria for utilizing an experimental combustion primer coating to optimize the ignitability of KNSB propellant grains. The primary objective of the primer is to act as an energetic thermal bridge, translating the transient flame front from a nichrome bridge wire or pyrogen igniter into uniform, simultaneous, multi-point grain ignition. This analysis details the exact chemical formulation, application protocols, geometric sizing constraints, and real-world ballistic recommendations for implementation in the project's upcoming subscale and M-class motor static firings.

2. PHYSICOCHEMICAL MECHANISMS AND THERMODYNAMIC MODELING

The enhanced ignitability achieved through a combustion primer relies on two primary physical and chemical phenomena: radiative heat transfer optimization and lowered thermal decomposition thresholds.

2.1. Radiative Heat Absorption via High Emissivity

Cast KNSB propellant is naturally translucent or white, resulting in a high surface albedo that reflects a significant portion of the radiant thermal energy emitted by the igniter pyrotechnic charge. By introducing a matte-black charcoal-rich primer coating, the surface emissivity (ε) is driven near unity ($\varepsilon \approx 0.95$). The instantaneous radiative heat flux (q_{rad}) absorbed by the grain surface is governed by the Stefan-Boltzmann law:

$$q_{rad} = \varepsilon \times \sigma \times (T_{flame}^{4} - T_{surface}^{4})$$

Where σ is the Stefan-Boltzmann constant ($5.6703 \times 10^{-8}\ W/(m^2 \cdot K^4)$), T_{flame} is the temperature of the igniter flame gases, and $T_{surface}$ is the local grain temperature. Maximizing ε directly accelerates the surface temperature rise time, bringing the boundary layer to activation parameters within milliseconds.

2.2. Thermal Decomposition Kinetics

The chemical mechanism of the 80/20 Potassium Nitrate (KNO_3) and Charcoal primer bypasses the sluggish decomposition of sorbitol. When exposed to the initial igniter trigger:

- **Phase 1 (Phase Change):** The KNO_3 component undergoes clean melting at $333°C$ (606.15 K). Because charcoal does not require a phase change or melt at this temperature, zero latent energy is wasted on fuel liquification.
- **Phase 2 (Exothermic Breakdown):** Upon reaching approximately $400°C$ (673.15 K), the molten KNO_3 vigorously decomposes, releasing highly reactive gaseous oxygen directly into the porous carbon matrix:

$$2KNO_3 \rightarrow 2KNO_2 + O_2\,(g)$$

  This rapid liberation of oxygen triggers the immediate, exothermic combustion of the embedded charcoal particles, generating a concentrated local heat flux exceeding $1000°C$. This high-intensity localized energy easily breaks the underlying KNSB auto-ignition boundary layer, achieving rapid flame spread.

3. PARAMETRIC PERFORMANCE DATA AND COMPARATIVE ANALYSIS

Empirical validation derived from historical propellant ignitability experiments demonstrates a striking disparity between bare and primer-coated grains. The time from initial flame exposure to full combustion initiation was measured across multiple standard 8-gram propellant slugs under controlled flame geometry.

Table 1: Baseline Ignitability of Uncoated Propellants (Time to Ignition in Seconds)

TEST NUMBER	KN-SUCROSE (KNSU)	KN-DEXTROSE (KNDX)	KN-SORBITOL (KNSB) - UNCOATED
Test 1	4.7 s	7.4 s	17.7 s
Test 2	5.9 s	8.0 s	22.0 s
Test 3	4.3 s	8.1 s	16.4 s
Test 4	4.8 s	7.2 s	22.8 s
Test 5	4.3 s	8.6 s	20.4 s
Mathematical Average	**4.8 s**	**7.9 s**	**19.9 s**

The baseline data shows that unprimed KNSB takes an average of 19.9 seconds to reach self-sustaining combustion—4.15 times longer than KNSU. This severe delay confirms the absolute necessity of a combustion primer for reliable motor operation.

Table 2: Enhanced Ignitability of KNSB Propellant via Combustion Primer Coating

TEST NUMBER	KN-SORBITOL (KNSB) WITH 80/20 PRIMER	IGNITION ACCELERATION FACTOR
Test 1	2.9 s	6.86 ×
Test 2	4.8 s	4.15 ×
Test 3	5.0 s	3.98 ×
Test 4	12.8 s (Aberrant Data Point)*	1.55 ×
Test 5	3.3 s	6.03 ×
Full Average (All Data)	**5.8 s**	**3.43 ×**
Optimized Average (Excl. Aberration)	**4.0 s**	**4.98 ×**

*Note: The aberrant 12.8-second data point in Test 4 is attributed to a localized binder failure or incomplete solvent evaporation during the curing cycle, emphasizing the critical nature of the application technique.

4. TECHNICAL FORMULATION AND APPLICATION PROTOCOL

To ensure reproducible ballistic performance and robust mechanical adhesion, the R&D team must strictly enforce the following chemical formulation and application procedures.

4.1. Chemical Composition and Solvent Carrier

The combustion primer formulation is standardized as follows:

- **Dry Mix Component:** 80% Potassium Nitrate (KNO_3) and 20% Charcoal by weight. The components must be processed together using a mortar and pestle until pulverized into a sub-micron, ultra-fine dust to ensure intimate chemical contact.
- **Volatile Solvent Vehicle:** 70% Isopropyl Alcohol (C_3H_8O / Rubbing Alcohol). The solvent is added incrementally to the dry powder while continuously grinding until the mixture achieves the smooth, homogenous consistency of thick liquid paint.

4.2. Application and Curing Methodology

The mixture must be applied using a high-density, fine-bristle paintbrush directly onto the designated propellant surfaces in a single, continuous, uniform coat. After application, the primed grains must undergo a mandatory curing period of a **minimum of 24 hours** in a dehumidified, desiccant-assisted environment. This allows complete evaporation of the volatile isopropyl alcohol carrier, leaving behind a highly durable, scratch-resistant, and tightly adhered dry energetic film.

5. SIZING CONSTRAINTS, QUANTITY OPTIMIZATION, AND FLUID DYNAMICS

Controlling the precise quantity and spatial distribution of the applied primer is vital to avoid catastrophic internal ballistics failure. The nominal dry film thickness (DFT) must target exactly **0.0075 inches (200 microns)**.

5.1. The Fluid Dynamics Stagnation Zone Problem

In standard BATES (Bifold Cylinder) grain configurations, the central core of the propellant segments experiences high-velocity, turbulent gas flow expelled by the central pyrogen igniter. This high convective heat transfer coefficient (h_c) facilitates rapid ignition along the core wall. However, the flat transverse ends of the BATES segments represent aerodynamic **stagnation zones** where gas flow velocities drop toward zero, forcing heat transfer to occur solely via weak conduction and radiation. Without a combustion primer, these segment ends suffer from extreme delayed ignition, causing the burning surface area—and consequently the motor's internal pressure profile—to deviate dangerously from design values.

5.2. Mass and Thickness Constraints

Deviating from the specified 200-micron thickness threshold introduces severe operational risks:

- **Insufficient Quantity (< 100 microns):** A coat that is too thin will lack the energetic mass required to overcome the local thermal inertia of the KNSB substrate, failing to sustain multi-point flame propagation across the stagnation zones.
- **Excessive Quantity (> 300 microns):** An overly thick coat acts as an insulating thermal barrier initially, slowing down ignition. Furthermore, because the 80/20 primer burns rapidly at ambient pressure with low gas volume yield, an excessive quantity can create an immediate, uncontrolled initial pressure spike upon ignition, or physically flake off under structural ignition shock, clogging the nozzle throat.

CATASTROPHIC HANGFIRE AND OVER-PRESSURIZATION HAZARD

Improper primer quantity or uneven coating thickness leads to asynchronous thermal breakdown. If one region of the grain ignites prematurely while the stagnation ends remain unignited, the localized pressure gradient can cause mechanical structural failure or segment cracking. Residual unevaporated solvent (isopropyl alcohol) acts as a powerful thermal sink, suppressing local temperatures and causing hazardous hangfires or complete ignition failure.

6. ACTIONABLE DESIGN AND ENGINEERING RECOMMENDATIONS

Based on this exhaustive physicochemical and parametric analysis, the following engineering standards are implemented for all future solid propulsion testing phases:

- **Standardize Primer Geometry:** Restrict combustion primer application to a single-layer coat with a verified dry film thickness of 200 microns (± 50 μm) on all flat segment ends of BATES grains and along the upper 25% of the internal core surface nearest the igniter head.

- **Mandatory Pre-Processing:** Enforce the use of pre-milled, air-float commercial charcoal and fully dehydrated, pulverized KNO_3. Avoid coarse granulated components, which introduce chemical inhomogeneities and cause uneven burn rates.
- **Environmental Curing Protocol:** All primed grains must cure for 24 hours at an ambient relative humidity (RH) of less than 30%. Grains must be weighed before and after coating to verify total dry solvent loss and accurately track the exact parasitic mass of the primer.
- **Ballistic Software Integration:** The total mass contribution of the dry primer film must be integrated into burn-simulation tools (such as OpenMotor). While the primer's primary purpose is ignition enhancement, its initial mass consumption must be modeled to ensure the transient startup pressure spike stays well within the safety margins of the aluminum motor casing.

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7. References

- Nakka, R., "Propellant Igniteability Experiment," Richard Nakka's Experimental Rocketry Web Site (https://www.nakka-rocketry.net/ignexp.html).
- NASA SP-8051, "Solid Rocket Motor Igniters," National Aeronautics and Space Administration Design Criteria Monograph.
- ASTM B344, "Standard Specification for Drawn or Rolled Nickel-Chromium Alloys for Electrical Heating Elements," ASTM International.